Pierpaolo **Vendittelli**

Postdoctoral Researcher · AI for Digital Pathology

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EXPERIENCE

Postdoc Researcher
Dec. 2024 – Present

Prinses Maxima Centre for Pediatric Oncology — *Utrecht, The Netherlands*

- Developing deep-learning and MIL pipelines for WSI analysis of Wilms tumor, targeting anaplasia grading and clinical outcome prediction.
- Coordinating multi-center dataset integration across collaborating hospitals; enforcing reproducible, clinically validated computational workflows.
- Translating model outputs into interpretable biomarkers in close collaboration with pathologists and pediatric oncologists.
- Supervising students and contributing to peer-reviewed publications in computational pathology and pediatric oncology.

PhD Candidate – AI for Digital Pathology
Mar. 2021 – Jun. 2026 (exp.)

Radboud University Medical Center — *Nijmegen, The Netherlands*

- Developed deep-learning segmentation models for pancreatic cancer WSIs within PANCAIM (EU Horizon 2020 consortium).
- Designed multi-class U-Net for automated TSR quantification across 368 patients / 4 datasets (25 collaborating NL labs); achieved Dice 0.75 (epithelium) and 0.86 (tumor bulk) on external TCGA-PAAD validation; TSR yielded AUC 0.61 for 6-month survival prediction on an independent cohort. Published in *PLOS ONE* (2024) and *SPIE MI* (2022).
- Teaching Assistant for *Image Analysis* and *Medical Imaging Analysis* across 4 course editions at Radboud University.
- Co-supervised Master's theses in computational medical imaging.

Machine Learning Engineer
Oct. 2020 – Feb. 2021

Everis Italy (NTT DATA) — *Rome, Italy*

- Built AI solutions on Google Cloud Platform for clients in banking, fashion, and automotive.
- Studied and applied GCP AI services (Vertex AI, BigQuery ML, AutoML) for custom model development and deployment.

Research Intern
Feb. 2020 – Sep. 2020

ViCOROB – NeuroImage Computing Group, University of Girona — *Girona, Spain*

- Benchmarked MRI synthesis architectures (ResU-Net, cGAN, CycleGAN) on WMH and BraTS 2018 datasets for T1↔FLAIR synthesis.
- Demonstrated ResU-Net achieves superior SSIM and lower MSE vs. GAN variants under data-scarce conditions; informed Master's thesis (grade 8.6/10).

PROJECTS & PUBLICATIONS

Weekly AI Paper-Triage Agent
Personal Project

github.com/PierpaoloV — *2024 – Present*

Multi-agent pipeline (OpenAI API, Python): weekly arXiv / PubMed scan → LLM summarisation → interest-based scoring (0–10) → automated email digest. In active personal use since 2024.

Automatic quantification of tumor-stroma ratio as a prognostic biomarker for pancreatic cancer
PLOS ONE

doi:10.1371/journal.pone.0301969 — *May 2024*

Multi-class U-Net for TSR segmentation across 368 patients / 4 datasets (25 NL labs). Dice 0.75 (epithelium) and 0.86 (tumor bulk) on external TCGA-PAAD validation. TSR AUC 0.61 for 6-month survival prediction on independent cohort.

Automatic quantification of TSR as a prognostic marker for pancreatic cancer
MIDL 2023

openreview.net/pdf?id=Dtz_iaUpGc — *Sep. 2023*

Poster presentation: segmentation of tumor epithelium, healthy epithelium and TSR estimation from pancreatic WSIs.

Setting the research agenda for clinical AI in pancreatic adenocarcinoma imaging
Cancers

doi:10.3390/cancers14143498 — *Jul. 2022*

Systematic review on the current role of AI for pancreatic cancer diagnosis and personalised treatment.

Automatic tumour segmentation in H&E-stained whole-slide images of the pancreas
SPIE MI 2022

doi:10.1117/12.2611542 — *Apr. 2022*

Multi-task U-Net with auxiliary classification head for coarse pancreatic tumour area segmentation in WSIs; classification head reduced false positives.

EDUCATION

Joint M.Sc. in Medical Imaging and Applications *Sep. 2018 - Sep. 2020*

University of Burgundy, University of Cassino and Southern Lazio, University of Girona *France, Italy, Spain*

- Final Grade: Bien (France), 102/110 (Italy), 8.6/10 (Spain)
- Scholarships: 2018-2020 Consortium Grant for the Erasmus Mundus Joint Master Degree.
- Relevant Courses: Computer-Aided Diagnosis, Medical Image Segmentation and Registration, Advanced Image Analysis.

B.Sc. in Informatics and Telecommunication Engineering *Sep. 2012 - Sep. 2017*

University of Cassino and Southern Lazio *Cassino, Italy*

- Final Grade: 94/110
- Scholarships: 2016 Erasmus+ Funded Exchange at Fontys University of Applied Sciences in Eindhoven, The Netherlands.
- Relevant Courses: Software Engineering, Databases, Mathematical Analysis, Object-Oriented Programming, Statistics.

Honors & Grants

2021 – 2024	**PhD Fellowship**, PANCAIM – Horizon 2020 EU-funded consortium for AI in pancreatic cancer	*Nijmegen, The Netherlands*
2018 – 2020	**Consortium Grant**, Erasmus Mundus Joint Master Degree in Medical Imaging and Applications (MAIA)	*France, Italy, Spain*
2016	**Erasmus+ Exchange Grant**, Funded exchange semester at Fontys University of Applied Sciences	*Eindhoven, The Netherlands*

Conference Presentations

Medical Imaging with Deep Learning (MIDL 2023) *Nashville, TN, USA*

Poster Presentation *Sep. 2023*

- "Automatic quantification of TSR as a prognostic marker for pancreatic cancer" – segmentation of tumor epithelium, healthy epithelium and tumor-stroma ratio from H&E WSIs.

SPIE Medical Imaging 2022 *San Diego, CA, USA*

Oral Presentation *Apr. 2022*

- "Automatic tumour segmentation in H&E-stained whole-slide images of the pancreas" – multi-task U-Net with classification head for pancreatic tumour area segmentation.

SKILLS

Languages: Python, C++, C, Bash, SQL
ML / Deep Learning: PyTorch, scikit-learn, Hugging Face Transformers, MONAI, Elastix, Keras
MLOps & Infrastructure: Docker, Git, GCP (Vertex AI, BigQuery ML), SLURM / GPU clusters, OpenSlide, W&B
LLM & Agents: OpenAI API, Anthropic API, RAG, prompt engineering, multi-agent orchestration

CERTIFICATIONS

- **Machine Learning**, Stanford University
- **Convolutional Neural Networks**, deeplearning.ai
- **Generative Adversarial Networks (GAN) Specialization**, deeplearning.ai
- **Improving Deep Neural Networks: Hyperparameter Tuning, Regularization & Optimization**, deeplearning.ai
- **Neural Networks and Deep Learning**, deeplearning.ai
- **AI for Medical Diagnosis**, deeplearning.ai
- **AI for Medical Prognosis**, deeplearning.ai
- **Applied Plotting, Charting & Data Representation in Python**, University of Michigan